

# Maschinelles Lernen (ML)
# Einführung

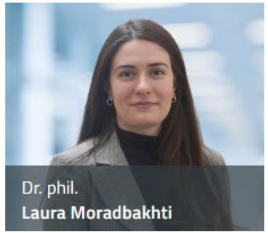
Dr.-Ing. Christoph-Alexander Holst

# Wer sind wir?

## Bildverarbeitung und Mustererkennung, Sensor- und Informationsfusion

### Gruppenleitung




### Seniorprofessor

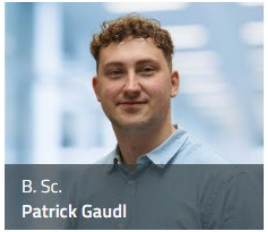


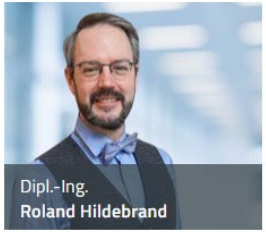

### Mitarbeiter







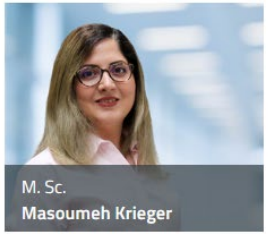




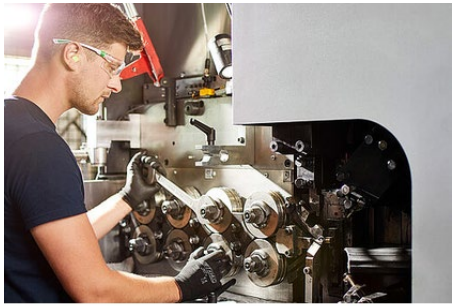







# Wer sind wir?



OptiCoil: KI-gestützte Prozessoptimierung für nachhaltige Stahlfederproduktion



ITS.ML: Intelligente Technische Systeme der nächsten Generation durch Maschinelles Lernen




itsowl-TP-DeepConcrete: Framework für die Kartierung von Stahlbetonfertigteilen aus Luftbildern für eine intelligente Außenlageroptimierung




PD Assist+: KI-gestützte Assistenz in der Parkinson-Versorgung

KI AKADEMIE OWL

# Wer sind wir?

**Übungsleitung**: Julian Knaup, M.Sc.

Übungen:

- Programmieren in Python
- Umsetzen verschiedener ML-Algorithmen „from scratch“
- Umgang mit den gängigsten ML-Bibliotheken (PyTorch, …)



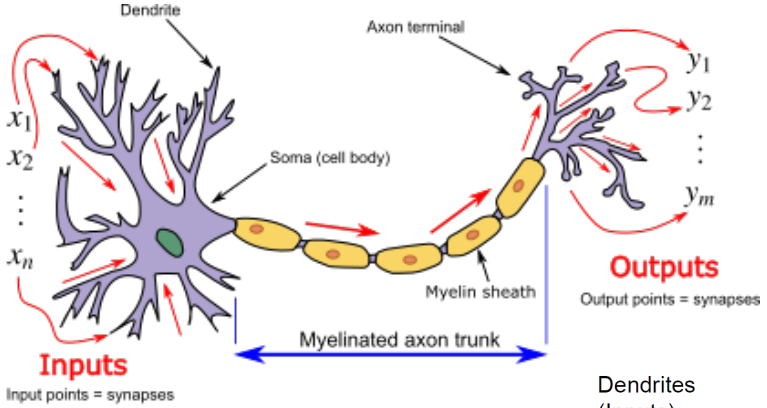
julian.knaup@th-owl.de

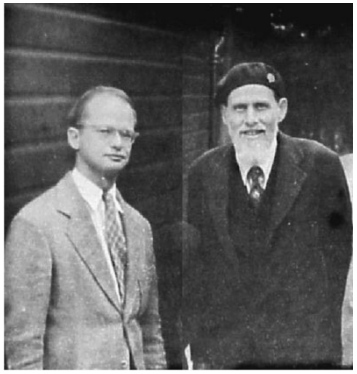
# Eine Historie des Maschinellen Lernens

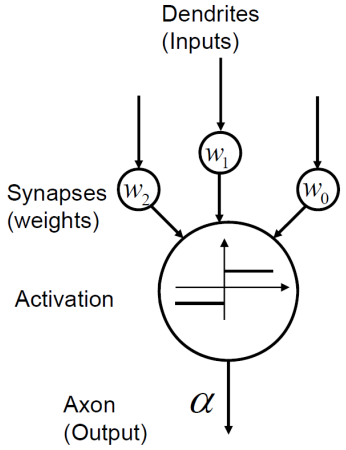
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McCulloch, Warren S., and Walter Pitts. "A logical calculus of the ideas immanent in nervous activity." *The bulletin of mathematical biophysics* 5.4 (1943): 115-133.

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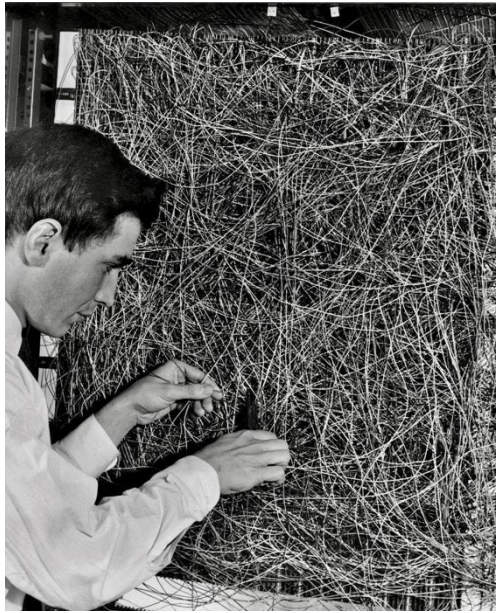

The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.

Wir lösen künstliche Intelligenz diesen Sommer.

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Frank Rosenblatt überführte McCulloch & Pitts Neuronen in single-layer und multi-layer Architekturen – das Perzeptron als erstes Neuronales Netz



Mark-I Perzeptron

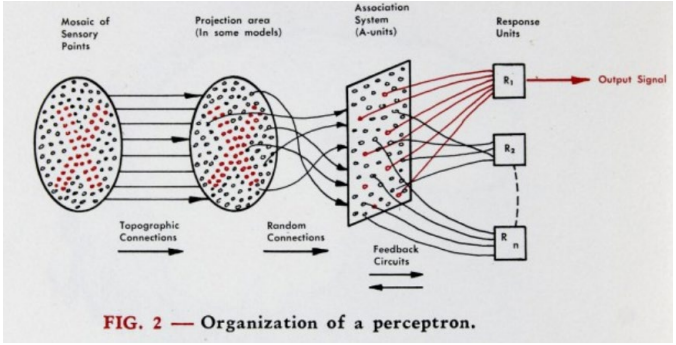



FIG. 2 — Organization of a perceptron.

Rosenblatt, Frank. "The perceptron: a probabilistic model for information storage and organization in the brain." *Psychological review* 65.6 (1958): 386.

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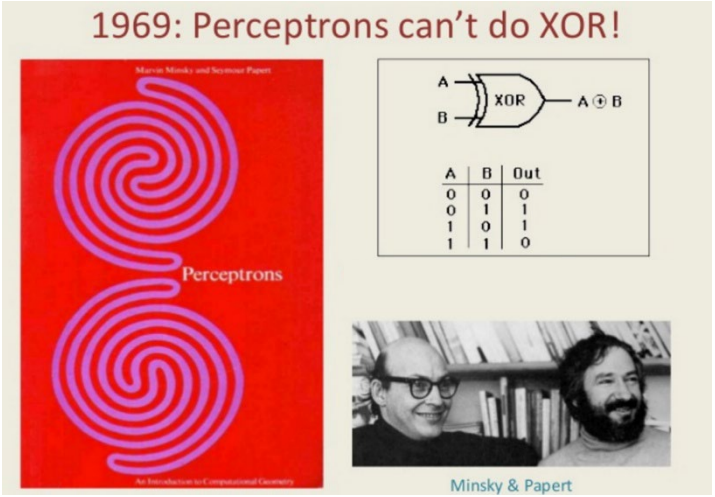
Samuel, Arthur L. "Some studies in machine learning using the game of checkers." *IBM Journal of research and development* 3.3 (1959): 210-229.

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**1966**: Maschinelle Übersetzung wird dem Hype nicht gerecht → Einsetzen des *Automatic Language Processing Advisory Committee (ALPAC)*

**1969**: Kritik am Perceptron durch Minsky und Papert [1]




**1973**: Lighthill Report [2] - vollständiger Abbau der KI-Forschung in den UK

[1] Minsky, Marvin, and Seymour A. Papert. *Perceptrons: An introduction to computational geometry*. the MIT Press, 2017.
[2] Lighthill, James. "Artificial intelligence: A general survey." *Artificial Intelligence: a paper symposium*. London: Science Research Council, 1973.

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**1980:** Expertensysteme treiben kommerziellen Boom. Z.B. XCON/R1 [1]

*Expertensystem:*

Menschl. Experte → Regeln → Logische Inferenz/Deduktion

*Maschinelles Lernen:*

Daten → Lernender Algorithmus → Modell

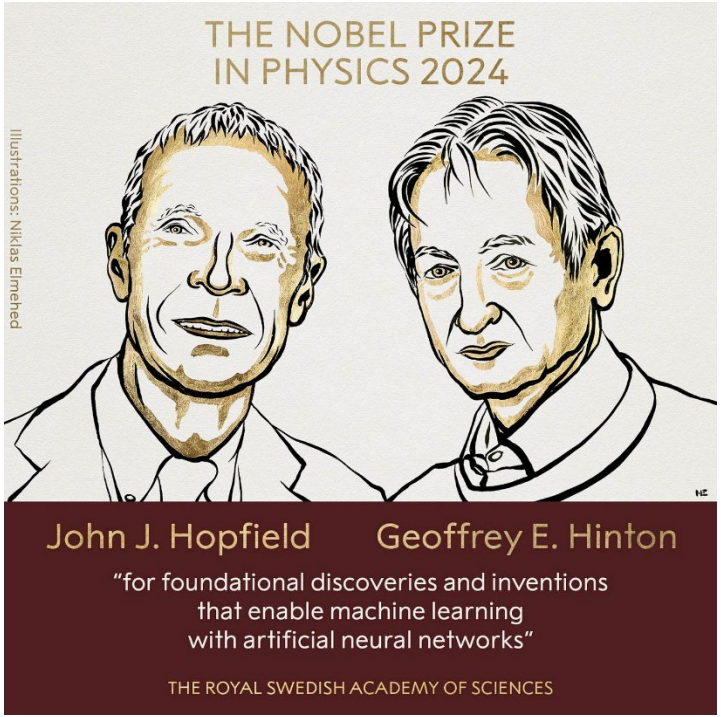



IBM Watson 2012(!)

[1] Polit, Stephen. "R1 and beyond: AI technology transfer at digital equipment corporation." *AI Magazine* 5.4 (1984): 76-76.

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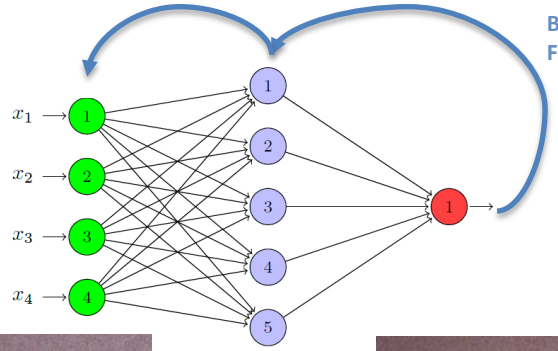


Hopfield, John J. "Neural networks and physical systems with emergent collective computational abilities." *Proceedings of the national academy of sciences* 79.8 (1982): 2554-2558.

Ackley, David H., Geoffrey E. Hinton, and Terrence J. Sejnowski. "A learning algorithm for Boltzmann machines." *Cognitive science* 9.1 (1985): 147-169.

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Werbos at IJCNN 1991



Rumelhart at IJCNN 1991


Rumelhart, David E., Geoffrey E. Hinton, and Ronald J. Williams. "Learning representations by back-propagating errors." *nature* 323.6088 (1986): 533-536.
Werbos, Paul J. "Backpropagation through time: what it does and how to do it." *Proceedings of the IEEE* 78.10 (1990): 1550-1560.

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Expertensysteme enttäuschen; Markt bricht ein; KI-Förderung wird erneut auf Eis gelegt.



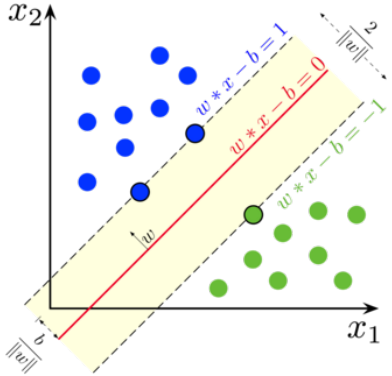
LISP Maschine

***Muster:***

Breakthrough → Hype → Unrealistische Erwartungen → Enttäuschung → Winter

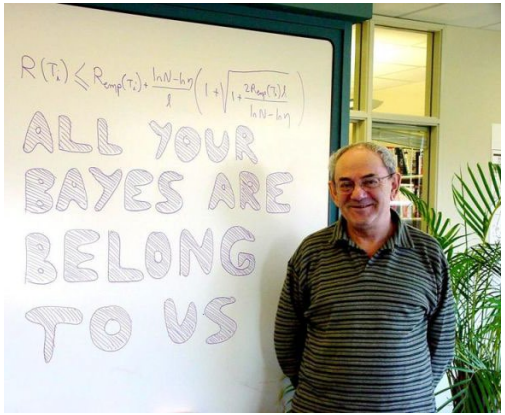
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Corinna Cortes




Vladimir Vapnik

Cortes, Corinna, and Vladimir Vapnik. "Support-vector networks." *Machine learning* 20.3 (1995): 273-297.

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Deep Blue


Garri Kasparov

Martina Heßler. Der Erfolg der Dummheit. Deep Blues Sieg über den Schachweltmeister Garri Kasparov und der Streit über seine Bedeutung für die Künstliche Intelligenz-Forschung. NTM Zeitschrift für Geschichte der Wissenschaften, Technik und Medizin, 25(1):1–33, 2017.



# Eine Historie des Maschinellen Lernens

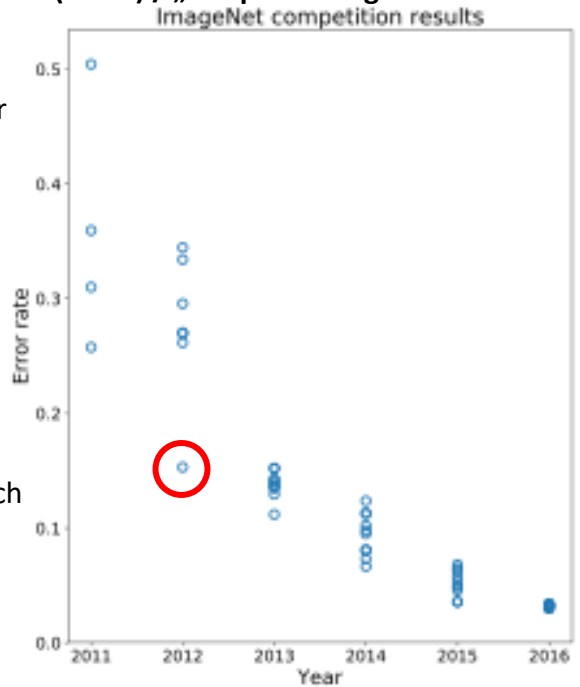
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**Convolutionale Neural Networks (CNNs) / „Deep Learning“**

**1980 – Fukushima: Neocognitron[1]**
Vorläufer moderner CNNs mit hierarchischer Bildrepräsentation.

**1990er – LeCun / LeNet[2]**
LeCun et al. dokumentieren 1998 produktive Anwendungen (u.a. Handschrift-/Dokumenterkennung)

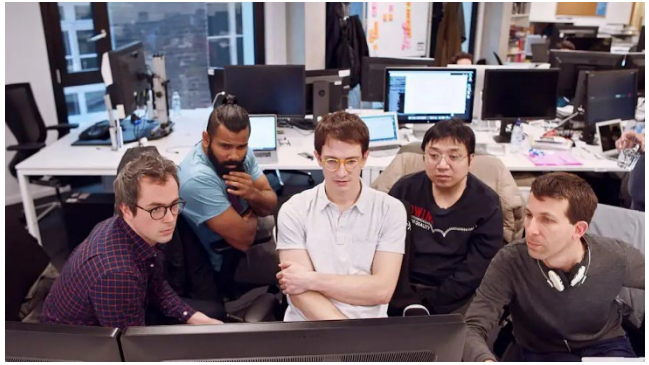
**2009/2012 – ImageNet[3] + AlexNet[4]**
ImageNet: riesige Bilderdatensätze
AlexNet schlägt damalige Konkurrenz deutlich




[1]Fukushima, Kunihiko. "Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position." *Biological cybernetics* 36.4 (1980): 193-202.
[2]LeCun, Yann, et al. "Gradient-based learning applied to document recognition." *Proceedings of the IEEE* 86.11 (1998): 2278-2324.
[3]Deng, Jia, et al. "ImageNet: A large-scale hierarchical image database." *2009 IEEE conference on computer vision and pattern recognition*. IEEE, 2009.
[4]Krizhevsky, Alex, Ilya Sutskever, and Geoffrey E. Hinton. "ImageNet classification with deep convolutional neural networks." *Advances in neural information processing systems* 25 (2012).

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Google DeepMind Team



Lee Sedol

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GPT: Generativ Pre-Trained Transformer

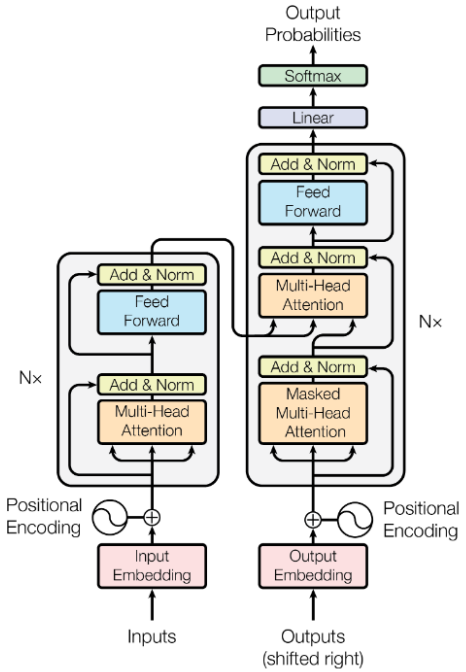
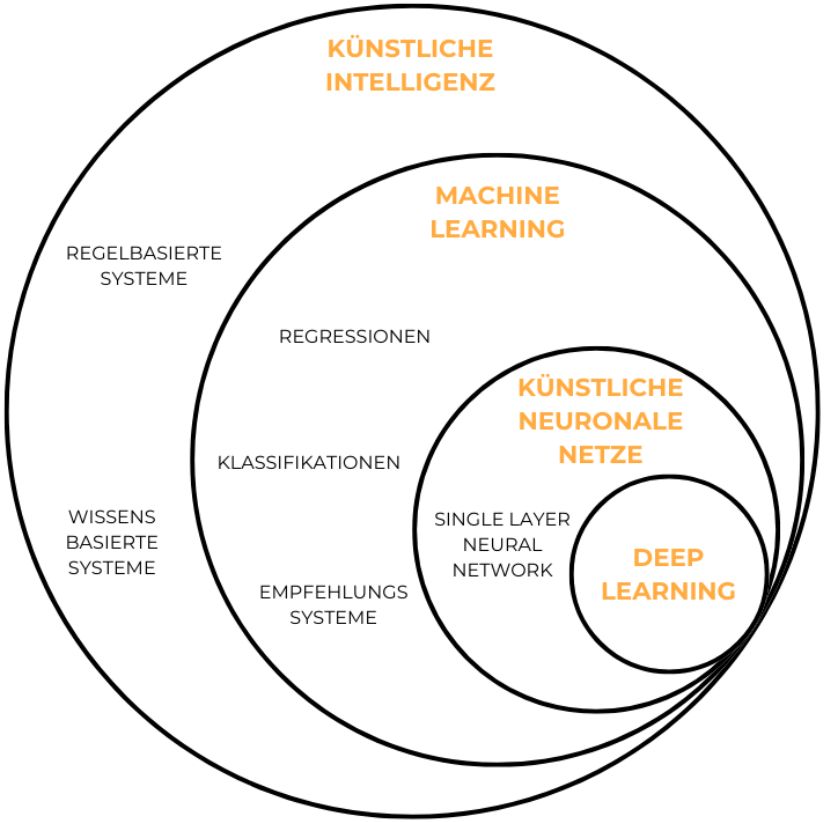



Figure 1: The Transformer - model architecture.

Vaswani, Ashish, et al. "Attention is all you need." Advances in neural information processing systems 30 (2017).

# Künstliche Intelligenz vs. Maschinelles Lernen vs. Deep Learning

$$Deep\ Learning \subset Maschinelles\ Lernen \subset Künstliche\ Intelligenz$$



# Was ist „Lernen“?

Mitchell, Tom M.,. *Machine learning*. Vol. 1. No. 9. New York: McGraw-hill, 1997.

# Was ist „Lernen“?

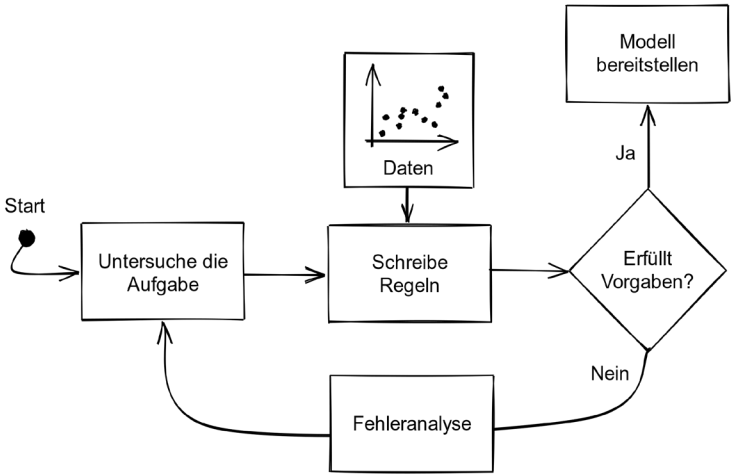
**Tom Mitchell**: „A computer program is said to learn from experience *E* with respect to some class of tasks *T* and performance measure *P*, if its performance at tasks in *T*, as measured by *P*, improves with experience *E*”

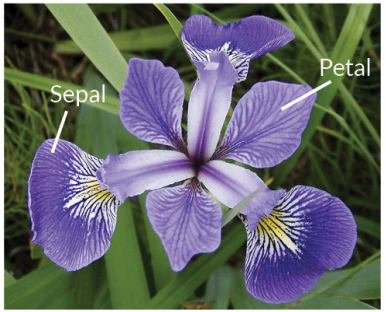
*Task*: Spam-Email erkennen
*Experience*: Durch Menschen vorkategorisierte Mails
*Performance*: Anteil der richtig eingeordneten Emails

Mitchell, Tom M.,. *Machine learning*. Vol. 1. No. 9. New York: McGraw-hill, 1997.

# Wie unterscheiden sich ML und DL?

Klassische Herangehensweise
(ohne Lernen)



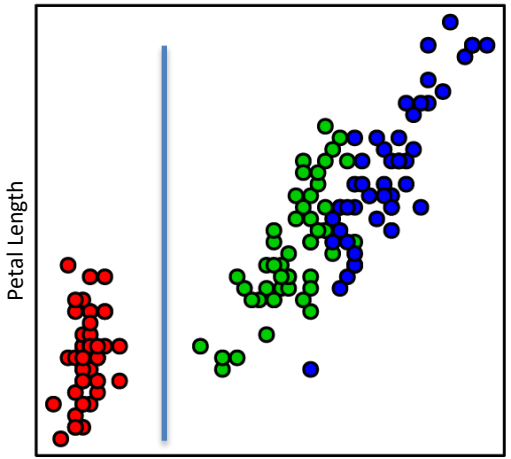

**Iris Versicolor**



**Iris Setosa**



**Iris Virginica**



# Wie unterscheiden sich ML und DL?

2014




https://xkcd.com/1425/

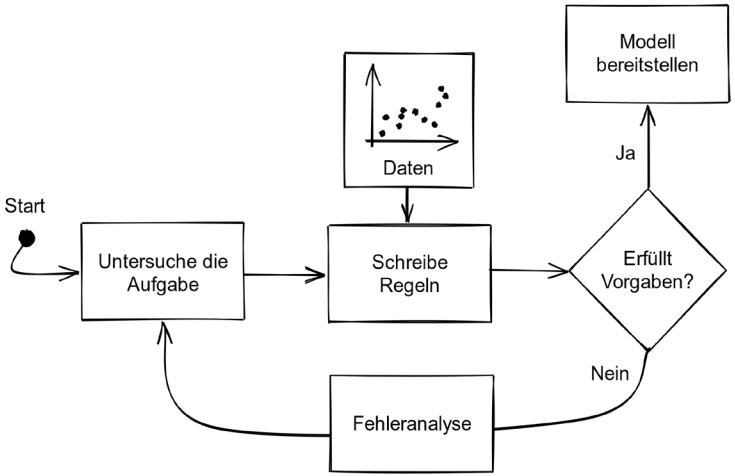
Polanyi-Paradox

„We know more than we can tell“



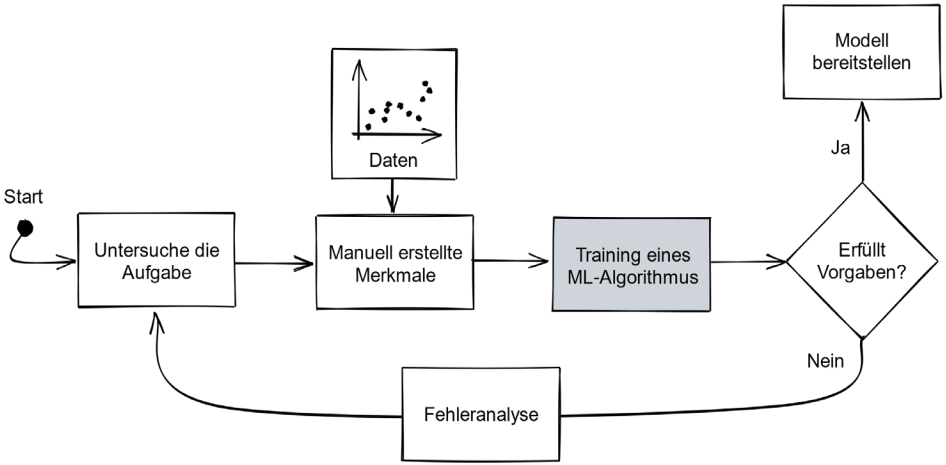
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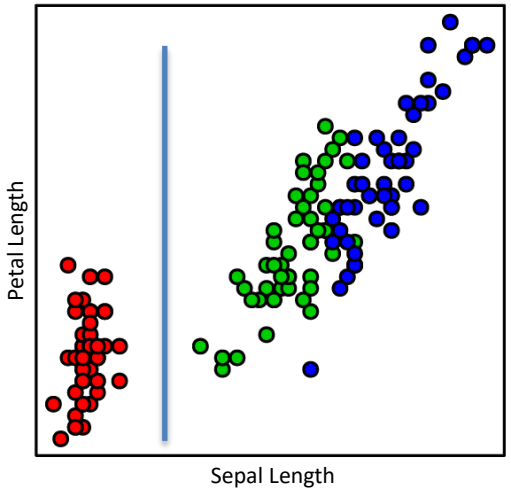
Klassische Herangehensweise
(ohne Lernen)



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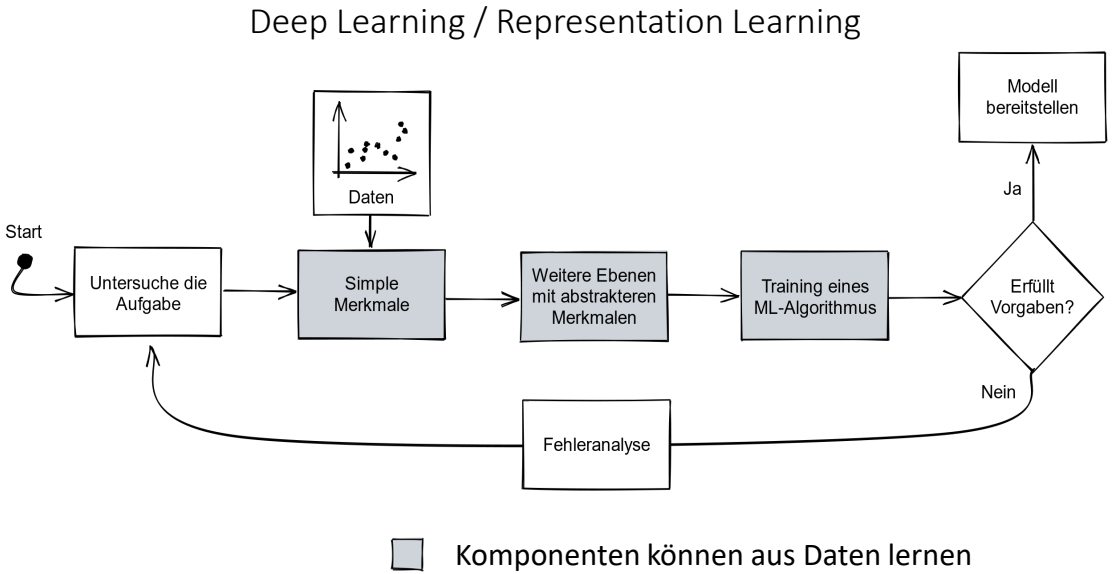
## Klassisches Maschinelles Lernen



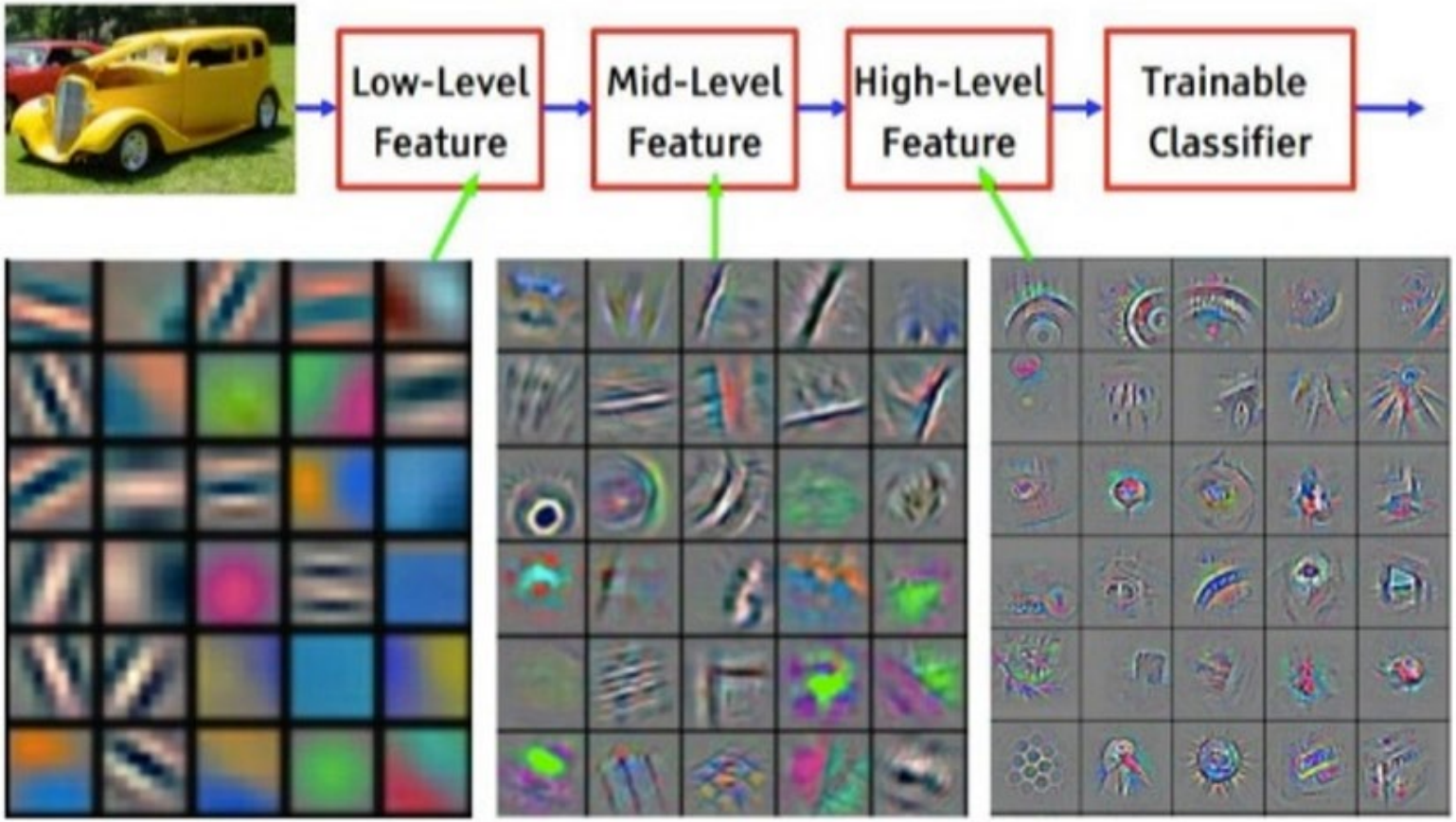




Komponenten können aus Daten lernen

# Wie unterscheiden sich ML und DL?

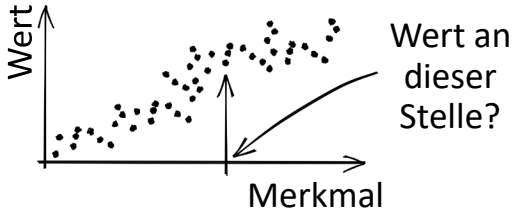


# Wie unterscheiden sich ML und DL?

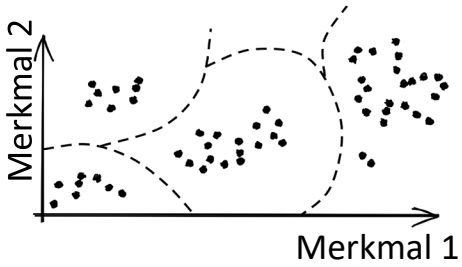


# Kategorisierung 1: Welche Art Problem lösen wir?

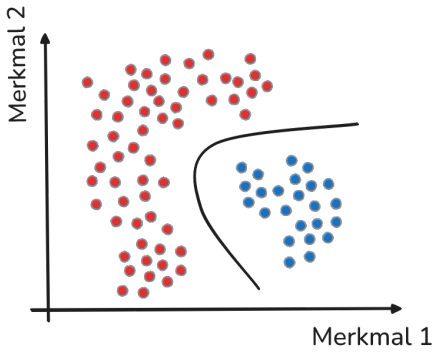
Regression




Clustering

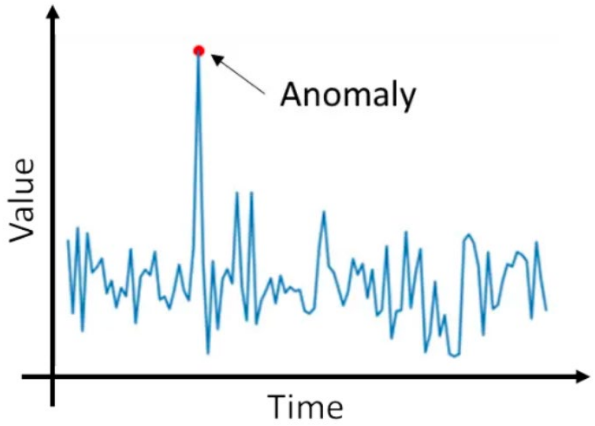



Klassifikation

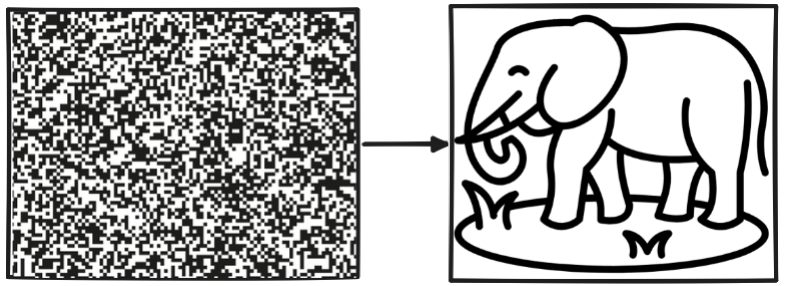


# Kategorisierung 1: Welche Art Problem lösen wir?

Anomalie Detektion




Generativ



# Kategorisierung 2: Woher kommt das Lernsignal?

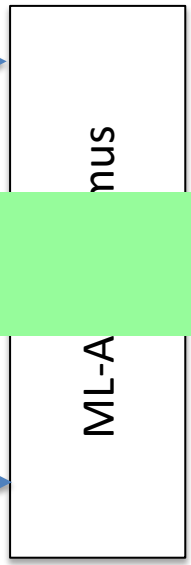
Überwachtes Lernen / Supervised Learning

Unüberwachtes Lernen / Unupervised Learning

Verstärkendes Lernen / Reinforcement Learning



„Katze“

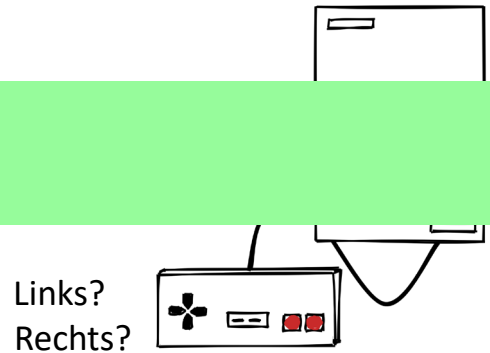


„Hund“



„Katze“



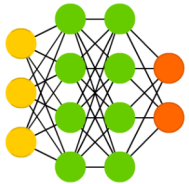


Links?
Rechts?

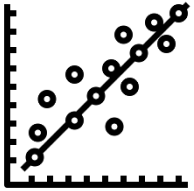
**+ Semi-Supervised Learning**
**Self-Supervised Learning**

# Kategorisierung 3: Mit welcher Modellfamilie?

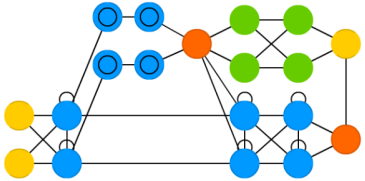
Neuronale Netze



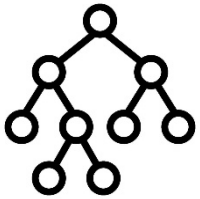
Lineare Modelle



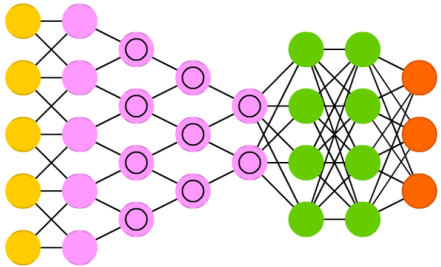
Transformer



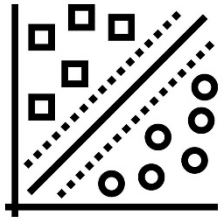
Trees / Random Forests

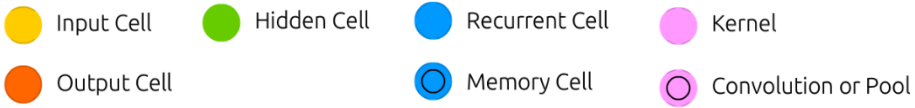


Convolutional Neural Networks



Support Vector Machines






https://www.asimovinstitute.org/neural-network-zoo/

| **Lernsignal** | **Aufgabe** | **Modell** |
|---|---|---|
| supervised | classification | Decision Tree |
| supervised | regression | Neural Network |
| unsupervised | clustering | k-Means |
| self-supervised | generation | Transformer |
| reinforcement | control/game playing | Neural Network |

# Zusammenfassung / Take Away

1. Hinter dem heutigen ML-Boom liegen Jahrzehnte Forschung – und mehrere Phasen überhöhter Erwartungen.
2. Maschinelles Lernen ersetzt explizite Regeln durch aus Daten gelernte Modelle.
3. KI, ML und DL sind nicht dasselbe.
4. „Supervised“, „unsupervised“ und „reinforcement“ beschreiben unterschiedliche Lernsettings.
5. Classification, Regression und Clustering sind Problemtypen – keine Algorithmen.

Nächster Termin: Wie lässt sich Lernen formal beschreiben?

# Vorlesungsstruktur

| Termin | Thema | Zweck |
|---|---|---|
| **01** | Was ist Machine Learning? | Was ist Lernen überhaupt? |
| **02** | Das Lernproblem: Daten, Repräsentation und Hypothesen | |
| **03** | Erste Modelle: Vom linearen Modell zum Perceptron | Ein erstes Modell bauen |
| **04** | Vom Perceptron zum neuronalen Netz | |
| **05** | Was bedeutet eigentlich „Lernen“? – Loss, Risk und Optimierung | Verstehen, was beim Lernen allgemein passiert |
| **06** | Generalisierung, Modellwahl und Evaluation | |

# Vorlesungsstruktur

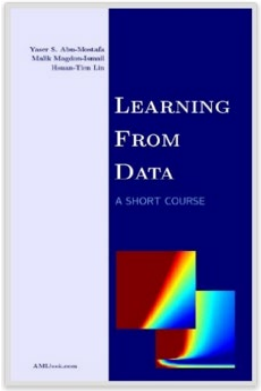
| Termin | Thema | Zweck |
|---|---|---|
| **07** | Decision Trees und Ensemble Learning | Andere Modell- und Lernfamilien kennenlernen |
| **08** | Support Vector Machines und Kernel Methods | |
| **09** | Unsupervised Learning I – Clustering | |
| **10** | Unsupervised Learning II – Repräsentation und Dimensionsreduktion | |
| **11** | CNNs – Lernen aus Bildern | Moderne Representation Learning-Systeme |
| **12** | Deep Learning in der Praxis / CNN II | |
| **13** | Moderne Lernparadigmen (Ausblick auf u.a. GenAI) | |

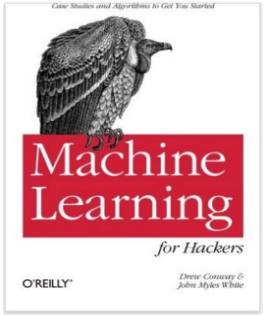
# Vorlesungsstruktur

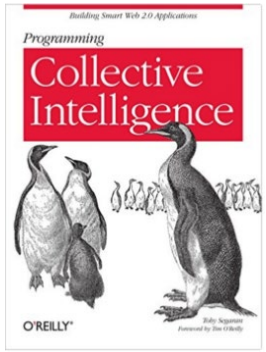
| Termin | Thema | Zweck |
|---|---|---|
| **14** | Wenn Machine Learning scheitert;<br>Verantwortungsvoller Einsatz | Grenzen verstehen |
| **15** | Puffer / Rückblick / Klausurvorbereitung | |

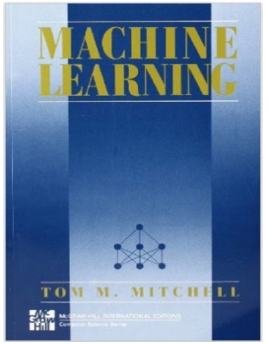
# Literatur zum Einstieg

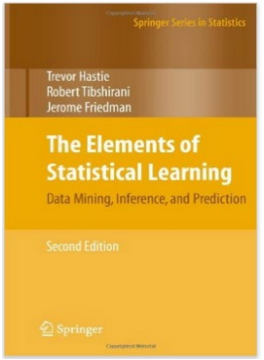
https://www.aisoma.de/maschinelles-lernen-und-datenanalyse-die-besten-5-buchempfehlungen-fuer-den-einstieg/



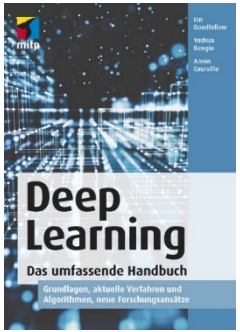



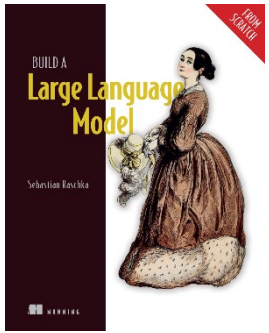



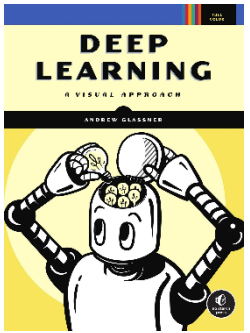
# Literatur zum Einstieg

Deep Learning Fokus

- Deep Learning by Ian Goodfellow, Yoshua Bengio and Aaron Courville (open access)
- Deep Learning - A visual Approach by Andrew Glassner
- Build a Large Language Model (from Scratch) by Sebastian Raschka





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